

Arun Soma

arunsoma0910@gmail.com

+1 (940) 882-8727

[LinkedIn](#) | [GitHub](#)



PROFESSIONAL SUMMARY

Full Stack Developer and AI Engineer with over 5 years of experience designing and delivering scalable, secure, and high-performance applications across Java, Go, Python, and C++. Skilled in building cloud-native microservices using Spring Boot, FastAPI, and event-driven architectures with Kafka, alongside modern frontend development with Angular and React Native. Proven track record in developing innovative AI solutions—including autonomous drones for precision agriculture and real-time sign language translation using deep learning models—combined with expertise in CI/CD automation, containerization, and multi-cloud infrastructure on AWS and GCP. Holds a Master's degree in Computer Science with a 4.0 GPA, excelling in Agile environments.

EDUCATION

Midwestern State University, Wichita Falls, TX

Master of Science, Computer Science GPA: 4.0/4.0

TECHNICAL SKILLS

- **Programming Languages:** Java, C#, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3
- **Frameworks & Libraries:** React.js, Node.js, Express.js, Angular, .NET Core, ASP.NET, Bootstrap, Material UI
- **Full-Stack Expertise:** MERN Stack (MongoDB, Express.js, React.js, Node.js)
- **Databases:** Microsoft SQL Server (2019), MongoDB, Mongoose, Azure Cosmos DB, MySQL
- **Cloud & DevOps:** Microsoft Azure (AKS, App Services, Service Bus, Event Hubs, Redis Cache), Docker, Kubernetes, Azure DevOps, CI/CD Pipelines, ARM/Bicep Templates
- **Authentication & Security:** JWT, OAuth2, Role-Based Access Control (RBAC), HTTPS/TLS, DevSecOps Practices
- **Architecture & Design:** Microservices, Event-Driven Architecture, RESTful API Design, MVC, Clean Architecture, SOLID Principles
- **Testing & Quality Assurance:** NUnit, Jest, Mocha, Postman, Swagger, Unit & Integration Testing, Continuous Testing
- **Tools & Collaboration:** Visual Studio, VS Code, Git, GitHub, Bitbucket, JIRA, Confluence, Agile/Scrum

PROJECTS

Precision Agriculture AI Drone System

- Built autonomous UAV system for weed detection and precision spraying.
- Trained a CNN on aerial crop images (98% test precision), deployed on drone platform.
- Developed real-time comms with Python sockets between laptop and onboard Jetson Nano.
- Published results in Springer Nature AIR-RES 2025

GestureAI – Real-Time Sign Language Translator

- Led development of a React Native application capturing ASL gestures and translating them to text in real-time using a custom PyTorch-trained video captioning model.
- Designed a CNN-LSTM architecture in PyTorch to process How2Sign video segments and generate tokenized sentence translations; trained using GPU acceleration with custom 'Dataset', 'DataLoader', and vocabulary tokenizer modules.
- Built a robust training pipeline with padding/collation, CrossEntropy loss, adaptive frame sampling (24 FPS), and checkpointing to save the best-performing model based on validation loss.
- Used mobile sockets to transmit gesture buffers to the backend inference server and return predictions to the app, ensuring minimal latency in real-time translation.

Twitter Sentiment Analysis for 2020 U.S. Election

- Applied VADER sentiment analysis on tweets with #DonaldTrump and #JoeBiden hashtags to classify public sentiment

- Aggregated and compared sentiment counts (Positive, Neutral, Negative) for both candidates; visualized results with matplotlib.
- Analyzed sentiment trends against official election outcomes to extract insights on public opinion dynamics.

Candy Store Mobile App (React Native)

- Built mobile app with screens for Search, Location, Chat, Image Gallery, and Auth.
- Developed backend in FastAPI and designed PostgreSQL schema. Firebase auth (in progress).

CERTIFICATION

- Certified AWS Developer – Associate.

PROFESSIONAL EXPERIENCE

Full Stack Java Developer

Aug 2024 – Present

Visa Inc, USA

- Followed full SDLC lifecycle in a fast-paced Agile Scrum setup using JIRA for sprint planning, backlog refinement, QA tracking, and Confluence for documenting service architecture, API integration workflows across distributed teams. Designed and developed scalable, secure Spring Boot microservices using Java 17 features, supporting tokenized payment processing and real-time transaction workflows.
- Implemented Kafka-based event streaming with Resilience4j for fault-tolerant, asynchronous inter-service communication across distributed payment systems.
- Built Angular 17 financial dashboards with reactive forms, NgRx state management, and optimized lazy loading, improving UI performance by 35%.
- Automated end-to-end CI/CD pipelines using Jenkins, GitHub Actions, Docker, and ArgoCD, accelerating deployment frequency and reliability on Kubernetes and AWS ECS.
- Architected cloud infrastructure with Terraform and AWS CloudFormation, enabling multi-cloud deployment across AWS and GCP for PCI-compliant financial services.
- Collaborated in Agile Scrum teams using JIRA and Confluence for sprint planning, documentation, and cross team API

Asa Bhanu Technical Services, Hyderabad, India

Dec 2022 – Aug 2023

Software Engineer

- Developed and optimized software solutions in C++ and MFC, reducing screen transition lag by 30%.
- Diagnosed startup animation lag caused by runtime 32-bit JPEG decoding; implemented pre-decoding and converted images to 24-bit BMP format, cutting fade-in time by over 40%.
- Enhanced UI/UX by restructuring interdependent menu systems into modular components using design patterns and SOLID principles, improving maintainability and preventing structural regressions during feature additions.
- Collaborated with cross-functional Agile teams to debug complex issues and produce comprehensive documentation.

Consultants to Government & Industry Inc., Bangalore, India

Jan 2020 – Nov 2022

Software Engineer

- Built and tested Go-based microservices for processing insurance user data, achieving over 90% unit test coverage using Go's native testing library; regularly validated peer modules to enhance team-wide productivity and accuracy.
- Diagnosed and resolved financial data rounding issues in a Red Hat Linux backend by identifying data structure limitations and updating C++ logic and MS SQL schemas to ensure precision in currency storage and calculations. Employed GNU Debugger and Linux process inspection tools to trace application behavior, pinpointing which backend processes were triggered by specific user calls and isolating the faulty module; mapped out inter-module dependencies and refactored them to improve maintainability.
- Optimized stored procedures and SQL functions, addressing over 80% of logged defects and improving overall system stability; verified all updates within a controlled sandbox environment.
- Collaborated with cross-functional teams to ensure robust integration between backend logic and database operations under strict compliance requirements.